Urine Specific Gravity (USG)

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Identity Sheet

Property	Value
Name	Urine Specific Gravity
Other names	USG, relative density, SG
Nature	Physical property (not a single molecule)
Definition	Ratio of urine density to water density at same temperature
Normal range	1.002-1.030
Typical value	1.015-1.025
Units	Dimensionless (g/mL relative to water = 1.000)

USG is a measure of the **concentration of dissolved solutes** in urine. It reflects the kidney's ability to concentrate or dilute urine. Major contributors: urea (~45%), NaCl (~25%), and other electrolytes and organic molecules. USG is a surrogate for osmolality but is influenced by the molecular weight of solutes (glucose and protein disproportionately raise USG).

Molecules Not to Be Confused With

Measure	Definition	Key Difference
Osmolality	Total solute particles per kg water (mOsm/kg)	True measure of concentrating ability; independent of MW; more accurate
Osmolarity	Solute particles per litre solution (mOsm/L)	Slightly different from osmolality; rarely used clinically
Total dissolved solids (TDS)	Mass of all dissolved substances	Related but measured differently

Medical Information

Origin

Endogenous USG reflects the net effect of all dissolved solutes in urine, which are products of metabolism (urea, creatinine, uric acid, electrolytes, organic acids, proteins). The kidney controls USG through the countercurrent mechanism and ADH-mediated water reabsorption in the collecting duct.

Exogenous Dietary intake of water and solutes directly affects USG. High water intake dilutes urine (low USG); high solute intake (protein, salt) concentrates urine (high USG). IV contrast media, mannitol, and dextran raise USG disproportionately.

Primary & Secondary Biological Roles

Primary role: - **Indicator of hydration and renal concentrating ability:** USG reflects the balance between water intake and renal water handling.

Secondary roles: - **Specimen adequacy assessment:** Used to determine if urine is too dilute for accurate testing of other analytes. - **Drug testing:** Specimens with USG <1.003 may be considered dilute/invalid.

Catabolism and Elimination Pathway

Not applicable – USG is a physical property, not a substance. It is determined by the concentrating and diluting functions of the nephron: ADH, countercurrent multiplication, urea recycling, and aquaporin-2 channels.

Expression in Humans

Normal Levels

Condition	USG Range
Normal (random)	1.002-1.030
First morning (concentrated)	1.015-1.030
After water loading	1.001-1.003
Maximum concentrating ability	1.030-1.040
Fixed (isosthenuria)	~1.010 (indicates renal concentrating defect)

Factors Influencing Levels **Increased USG (concentrated):** - Dehydration - SIADH - Adrenal insufficiency - Glycosuria (diabetes mellitus) - Proteinuria (nephrotic syndrome) - IV contrast media, mannitol - High protein diet

Decreased USG (dilute): - Excessive water intake (polydipsia) - Diabetes insipidus (central or nephrogenic) - Diuretic use - Chronic renal failure (loss of concentrating ability) - Hypercalcaemia, hypokalaemia

Associated Pathologies

Condition	USG Pattern	Key Symptoms
Diabetes insipidus	Persistently low USG (<1.005) despite dehydration	Polyuria, polydipsia
SIADH	Inappropriately high USG (>1.020) with hyponatraemia	Confusion, seizures
CKD (advanced)	Fixed USG ~1.010 (isosthenuria)	Nocturia, polyuria
Dehydration	High USG (>1.025)	Thirst, tachycardia
Water intoxication	Very low USG (<1.003)	Hyponatraemia, confusion

Presence in Urine

Should it be normally present? USG is always measurable. Normal range: 1.002-1.030.

Normal urinary levels: 1.015-1.025 (typical random specimen).

Form: Physical property of the entire urine specimen.

Pathological significance:

USG	Possible Causes	Prevalence
<1.005 (hyposthenuric)	DI, water loading, CKD	DI: ~1:25,000
1.010 (isosthenuric)	CKD, fixed dilution	CKD: ~10% of adults
>1.030	Dehydration, glycosuria, contrast	Common

Detection in Urine

Available Clinical Assays

1. **Refractometry (gold standard):**
   - **Principle:** Refractive index of urine is proportional to total dissolved solutes. Measured by critical angle refractometer.
   - **Detection:** Refractive index reading converted to SG scale.
   - **LOD:** Resolution ~0.001 SG unit.
   - **Advantages:** Accurate, rapid (<30 s), small volume needed.
   - **Disadvantages:** Affected by glucose and protein; temperature-dependent.
2. **Dipstick (ionic specific gravity):**
   - **Principle:** Polyelectrolyte (polymethylvinyl ether/maleic acid) in pad releases H^+ proportional to ionic strength -> pH indicator (bromothymol blue) colour change.
   - **Detection:** Colour change from blue-green (low SG) to yellow-green (high SG).
   - **LOD:** Reads 1.000 to 1.030 in 0.005 increments.
   - **Advantages:** Rapid, included on standard dipstick.
   - **Disadvantages:** Measures ionic strength only; does not detect non-ionic solutes (glucose, urea); alkaline pH can cause false low readings.
3. **Urinometry / hydrometry (historical):**
   - **Principle:** Floating hydrometer in urine; buoyancy proportional to density.
   - **Detection:** Direct reading of density scale.
   - **LOD:** ~0.001.
   - **Advantages:** Simple, no reagent.
   - **Disadvantages:** Large volume needed (15-20 mL); largely obsolete.
4. **Oscillating U-tube densitometry:**
   - **Principle:** Resonant frequency of a U-tube filled with urine depends on fluid density.
   - **Detection:** Frequency measurement.
   - **LOD:** ~0.0001 g/mL.
   - **Advantages:** Very precise; automated.

Optimal Urine Type for Measurement

- **First morning specimen** for assessing concentrating ability.
- **Random specimen** adequate for routine hydration assessment.
- Water deprivation test: serial USG measurements to assess concentrating capacity.
- Temperature correction needed for refractometry (standardised to 20 C).

Actual Gold Standard

Refractometry is the reference clinical method. Digital refractometers (Reichert, Atago) provide accurate, temperature-compensated readings. Precision: +/-0.001.

Interferences in Measurement

Interference	Effect	Mechanism
Glucose (>1 g/dL)	Falsely high SG (refractometer)	High-MW solute; dipstick unaffected (non-ionic)
Protein (>1 g/dL)	Falsely high SG (refractometer)	High-MW solute
IV contrast media	Very high SG (>1.040)	Dense iodinated molecules
Alkaline pH (>8)	Falsely low SG (dipstick)	Interferes with polyelectrolyte reaction
Temperature	Variable (refractometer)	Must correct to 20 C standard

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **NIR spectroscopy:** Urine water content and solute concentration affect NIR absorption (O-H overtone bands ~1450, 1940 nm). Multivariate calibration can predict USG.
  - LOD: Resolution ~0.002 SG units.
  - Sample prep: None (direct measurement).
- **UV-Vis:** Not directly applicable for a bulk physical property.

Fluorescence Detection

- Not applicable – USG is a physical property, not a fluorescent analyte.

Raman Detection

- **Water O-H stretch (~3400 cm-1):** Raman intensity of water band inversely correlates with solute concentration.
- Potentially usable for USG estimation via chemometric models.
- Not validated for clinical use.

FTIR Detection

- Water absorption bands dominate the MIR spectrum. The balance of water vs solute bands correlates with USG.
- ATR-FTIR with PLS calibration can estimate USG from overall spectral profile.
- LOD: ~0.003 SG units.
- Sample prep: None (direct ATR measurement).

Voltammetry Detection

- **Conductivity measurement:** Ionic strength (which partially correlates with USG) can be measured by conductivity electrodes.
  - LOD: ~0.1 mS/cm.
  - Correlates with dipstick ionic SG.
  - Does not measure non-ionic solutes.

Other Detection Technologies

1. **Acoustic resonance densitometry:** Ultrasonic speed-of-sound in urine correlates with density. LOD: ~0.0001 g/mL.
2. **Microfluidic density sensors:** MEMS-based resonant sensors. Prototype stage.
3. **Smartphone refractometer:** Optical attachments for phone cameras. LOD: ~0.002.

Summary Table: Detection Methods Comparison

Method	LOD/Resolution	Key Parameter	Sample Prep	Strengths
Refractometry	~0.001	Refractive index	None	Gold standard, fast
Dipstick (ionic SG)	~0.005	pH indicator colour	None	Included on dipstick
Oscillating U-tube	~0.0001	Resonant frequency	None	Very precise
NIR spectroscopy	~0.002	1450, 1940 nm	None	Reagent-free
ATR-FTIR	~0.003	Overall spectrum	None	Multianalyte
Conductivity	N/A	mS/cm	None	Real-time
Acoustic densitometry	~0.0001	Speed of sound	None	Non-contact

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References

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